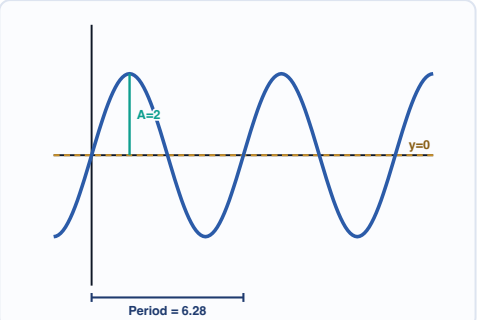
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

The Laplace Transform, Step Forcing & Convolution Practice Exercises

1 MCQ

Find $\mathrm{L}\{3t + \cos(t)\}$.



A $\frac{2s^3+3s^2+s+3}{s^4+s^2}$

B $\frac{-s^3-3s^2-3}{s^4+s^2}$

C $\frac{s^3+3s^2+3}{s^3+s}$

D $\frac{s^3+3s^2+3}{s^4+s^2}$

2 MCQ

Using the definition, $\mathrm{L}\{e^{2t}\} = \int_0^\infty e^{-(s-2)t}\,dt$ equals (for $s > 2$):

A $\frac{s}{s-2}$

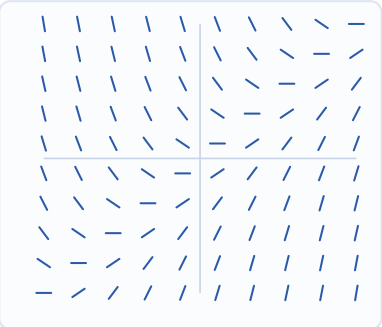
B $\frac{1}{s-2}$

C $\frac{1}{s+2}$

D $-\frac{1}{s-2}$

3 MCQ

If $\mathrm{L}\{y\} = Y(s)$, $y(0) = -3$ and $y'(0) = -1$, find $\mathrm{L}\{y''\}$.



A $Y s^2 + s + 3$

B $Y s^2 - 3s - 1$

C $Y s^2 + 3s + 1$

D $Y s + 4$

4 MCQ

Find $\mathrm{L}\{t^4\}$.

A $\frac{24}{s^4}$

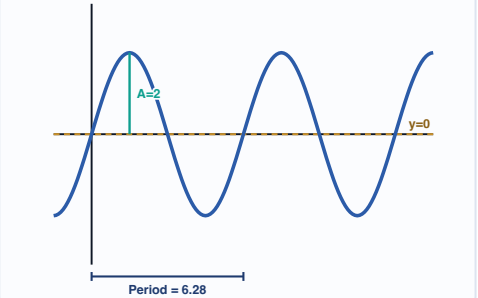
B $\frac{24}{s^5}$

C $\frac{4}{s^5}$

D $\frac{24}{s^6}$

5 MCQ

Using the definition, $\mathrm{L}\{2t\} = \int_0^\infty 2te^{-st}\,dt$ equals (for $s > 0$):



A $\frac{2}{s}$

B $\frac{2}{s^3}$

C $\frac{2}{s^2}$

D $-\frac{2}{s^2}$

6 MCQ

The second-order linear homogeneous ODE $f''(x) - 53f'(x) + 696f(x) = 0$ has characteristic roots λ_1, λ_2. What is the trace sum $\lambda_1 + \lambda_2$?

A 53

B 56

C 77

D 28

7 MCQ

If $\mathrm{L}\{y\} = Y(s)$ and $y(0) = 1$, find $\mathrm{L}\{y'\}$.

A $Ys - 1$

B $Ys + 1$

C $Y - 1$

D Ys

8 MCQ

Find $\mathrm{L}\{t^3\}$.

A $\frac{6}{s^4}$

B $\frac{6}{s^3}$

C $\frac{3}{s^4}$

D $\frac{6}{s^5}$

9 MCQ

Consider the differential equation $y''(t) + 4y(t) = u(t-2)$, where $u(t)$ is the Heaviside step function, with initial conditions $y(0) = 0$ and $y'(0) = 0$. Apply the Laplace Transform to both sides of the equation. If $Y(s) = \mathrm{L}\{y(t)\}$ and $\mathrm{L}\{u(t-a)\} = \frac{e^{-as}}{s}$, which of the following expressions represents $Y(s)$?

A $\frac{1}{(s^2+4)}$

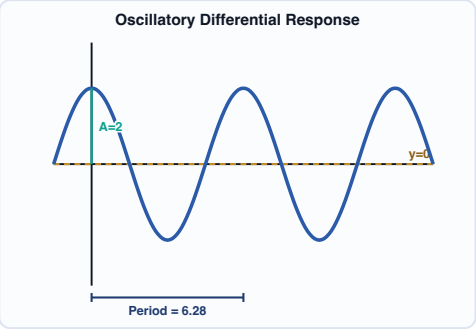
B $\frac{1}{s(s^2+4)}$

C $\frac{e^{-2s}}{s^2+4}$

D $\frac{e^{-2s}}{s(s^2+4)}$

10 MCQ

The second-order linear homogeneous ODE $f''(x) - 39f'(x) + 350f(x) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.



A 24

B 39

C 42

D 53

11 MCQ

Consider the initial value problem $y'' + 4y = \delta(t-\pi)$ with $y(0) = 0$ and $y'(0) = 0$. Which of the following represents the Laplace Transform of the solution, $Y(s) = \mathrm{L}\{y(t)\}$?

A $\frac{e^{-\pi s}}{s^2+4}$

B $\frac{1}{s^2+4}$

C $\frac{e^{-\pi s}}{s^2}$

D $\frac{e^{-\pi s}}{s^2-4}$

12 MCQ

The separable initial value problem $\frac{df}{dx} = 40x$ satisfies $f(0) = 4$. Determine the solution value $f(16)$.

A 5144

B 5120

C 5124

D 5127

13 MCQ

A damped spring satisfies $y'' + 2y' + 10y = 0$, $y(0) = 0$, $y'(0) = 6$. Find $y(\pi/6)$.

A $2e^{\pi/6}$

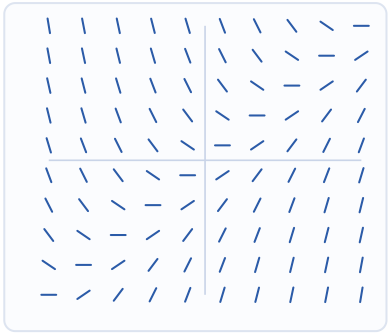
B $2e^{-\pi/6}$

C $-2e^{-\pi/6}$

D $e^{-\pi/6}$

14 MCQ

Solve $y'' + 4y' + 4y = e^{-2x}$, $y(0) = 0$, $y'(0) = 0$. Find $y(2)$.



A $2e^4$

B e^{-4}

C $4e^{-4}$

D $2e^{-4}$

15 MCQ

Calculate the solution set for $y'' - 7y' + 12y = 0$, $y(0) = 0$, $y'(0) = 3$. Find $y(1)$.

A $3e^3 + 3e^4$

B $-3e^3 + 3e^4$

C $-3e^4 + 3e^3$

D $-6e^3 + 6e^4$

16 MCQ

Solve $y'' + y' - 2y = 0$, $y(0) = 0$, $y'(0) = -6$. Find $y(1)$.

A $-2e + \frac{2}{e^2}$

B $-4e + \frac{4}{e^2}$

C $-2e - \frac{2}{e^2}$

D $-\frac{2}{e^2} + 2e$

17 MCQ

Find the root(s) satisfying $y'' - 2y' + y = 0$, $y(0) = 1$, $y'(0) = 2$. Find $y(1)$.

A 0

B $2e$

C $-2e$

D $\frac{2}{e}$

18 MCQ

Solve $y'' + 9y = 0$, $y(0) = -3$, $y'(0) = 6$. Find $y(\frac{\pi}{3})$.

A -2

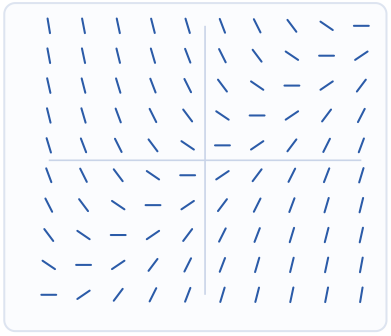
B -3

C 6

D 3

19 MCQ

Calculate the solution set for $y'' + 4y = 0$, $y(0) = -2$, $y'(0) = 6$. Find $y(\frac{\pi}{2})$.



A 4

B -3

C -2

D 2

20 MCQ

Solve $y'' - 4y' + 3y = 0$, $y(0) = -5$, $y'(0) = -7$. Find $y(1)$.

A $-4e^3 - e$

B $-e^3 - 4e$

C $4e + e^3$

D $-4e + e^3$

21 MCQ

Find the root(s) satisfying $y'' + 3y' + 2y = 0$, $y(0) = 5$, $y'(0) = -7$. Find $y(2)$.

A $-\frac{3}{e^2} - \frac{2}{e^4}$

B $-\frac{3}{e^2} + \frac{2}{e^4}$

C $\frac{2}{e^4} + \frac{3}{e^2}$

D $\frac{3}{e^4} + \frac{2}{e^2}$

22 MCQ

Determine the solution to $y'' - 6y' + 9y = 0$, $y(0) = 2$, $y'(0) = 8$. Find $y(2)$.

A $-6e^6$

B $\frac{6}{e^6}$

C $6e^6$

D $-2e^6$

23 MCQ

Work out the solution to $y'' + 6y' + 9y = 0$, $y(0) = -2$, $y'(0) = 3$. Find $y(1)$.

A e^{-3}

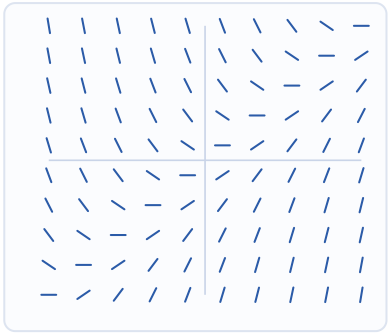
B $\frac{5}{e^3}$

C $-5e^3$

D $-\frac{5}{e^3}$

24 MCQ

Solve $y'' - 4y' + 8y = 0$, $y(0) = -1$, $y'(0) = 2$. Find $y(\frac{\pi}{6})$.



A $-\sqrt{3}e^{\frac{\pi}{3}} - \frac{e^{\frac{\pi}{3}}}{2}$

B $-\frac{\sqrt{3}e^{\frac{\pi}{3}}}{2} + e^{\frac{\pi}{3}}$

C $-\sqrt{3}e^{\frac{\pi}{3}} + \frac{e^{\frac{\pi}{3}}}{2}$

D $-\frac{e^{\frac{\pi}{3}}}{2} + \sqrt{3}e^{\frac{\pi}{3}}$

25 MCQ

Work out the solution to $y'' - 12y' + 35y = 0$, $y(0) = -1$, $y'(0) = -13$. Find $y(2)$.

A $3e^{10} + 4e^{14}$

B $-4e^{10} + 3e^{14}$

C $-3e^{10} + 4e^{14}$

D $-4e^{14} + 3e^{10}$